

Lecture 4

- P : a sphere packing w./ same radii, non-overlapping in $\mathbb{R}^d$

Def Sphere packing density

$$\theta(d) = \sup_P \limsup_{R \rightarrow \infty} \frac{\text{vol}(P \cap B_R(0))}{\text{vol}(B_R(0))}$$

- For high dim (d large)

Upp bd

- Kabatiansky - Levenshtein $\theta(d) \leq 2^{-0.599d}$

- Cohn - Zhao: constant factor improvement

Lower bd

- Exer $\theta(d) \geq 2^{-d}$

Hint: • consider a maximal packing

- recall $\text{vol}(B_R(0)) \sim \frac{1}{\sqrt{d\pi}} \left(\frac{2\pi e}{d}\right)^{d/2} R^d$

Venkatash

- for large d , $\theta(d) \geq 65963 \cdot d \cdot 2^{-d}$
- along a sparse seq of dim $\{d_i\}$:

$$\theta(d_i) = \Omega(d_i \log \log d_i \cdot 2^{-d_i})$$

Recall Hard-core model (over $\mathcal{Z}(G)$)

Partition function $Z_G(\lambda) = \sum_{k \geq 0} \underline{i_k(G)} \cdot \lambda^k$

uniform over each layer

k^{th} layer is weighted by fugacity

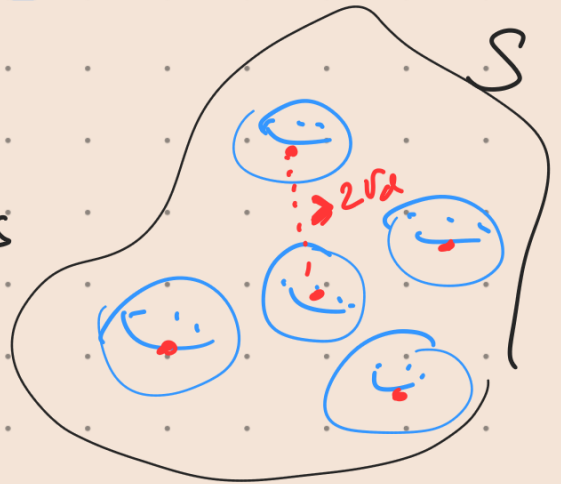
Def Hard-sphere model over bounded measurable set $S \subseteq \mathbb{R}^d$

Partition function $Z_S(\lambda) = \sum_{k \geq 0} \lambda^k \hat{Z}_S(k)$

where

$\hat{Z}_S(k) = \text{wt. of size-}k \text{ packings}$

$$= \frac{1}{k!} \int_S \mathbb{1}_{D(x_1, \dots, x_k)} dx_1 \dots dx_k$$



$D(x_1, \dots, x_k) = \text{event that } \forall i \neq j \quad d(x_i, x_j) > 2r_d$

• $\hat{Z}_S(0) = 1$

• Sample $X \sim \text{Hard-sphere dist. over } S$

$$P(|X| = k) = \frac{\lambda^k \hat{Z}_S(k)}{Z_S(\lambda)}$$

Def: expected packing density

$X \sim$ Hard sphere over S

$$\alpha_S(\lambda) = \frac{\mathbb{E}_{S,\lambda} |X|}{\text{vol}(S)}$$

Thm (Jenssen - Joos - Perkins) S bdd meas. $\in \mathbb{R}^d$

$$\forall \lambda \geq 3^{-d/2}$$

$$\theta(d) \geq \alpha_S(\lambda) \geq (1+o(1)) \cdot \log\left(\frac{2}{\sqrt{3}}\right) \cdot d \cdot 2^{-d}$$

≈ 0.1438

a factor of 2.4

Thm (Gil-Fernández, Kim, Liu, Pikhurko 21+)

$\forall \varepsilon > 0, \exists \delta > 0$ and d_0 s.t. $\forall d > d_0$

$$\forall \lambda \geq \left(\frac{1}{\sqrt{2}} - \delta\right)^d \Rightarrow \alpha_S(\lambda) \geq (\log \sqrt{2} - \varepsilon) \cdot d \cdot 2^{-d}$$

≈ 0.3465

Basic properties

Lem $\alpha_S(\lambda)$ is the scaled log. derivative of the

partition,
$$\alpha_S(\lambda) = \frac{\lambda}{\text{vol}(S)} \cdot \left(\log Z_S(\lambda) \right)'$$

Pf:
$$\begin{aligned} \alpha_S(\lambda) &= \frac{\mathbb{E}|X|}{\text{vol}(S)} = \frac{1}{\text{vol}(S)} \sum_{k=1}^{\infty} k \cdot \mathbb{P}(|X|=k) \\ &= \frac{1}{\text{vol}(S)} \sum_{k=1}^{\infty} k \cdot \frac{\lambda^k Z_S(k)}{Z_S(\lambda)} \end{aligned}$$

$$\therefore = \frac{\lambda}{\text{vol}(S)} \frac{Z'_S(\lambda)}{Z_S(\lambda)} = \frac{\lambda}{\text{vol}(S)} \left(\log Z_S(\lambda) \right)' \quad \square$$

Lem $\alpha_S(\lambda)$ is monotone increasing w.r.t. λ .

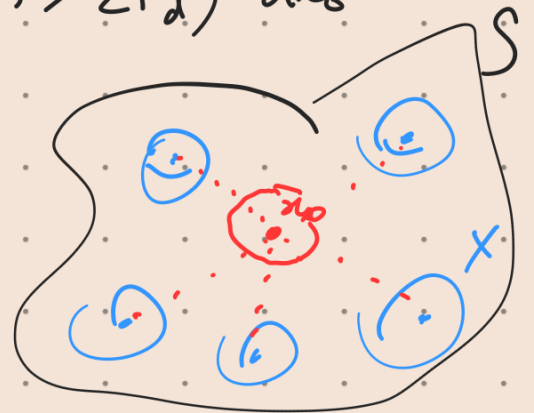
Pf: Use previous lem that $\alpha_S(\lambda) = \frac{\lambda}{\text{vol}(S)} \left(\log \right)'$

$$\lambda \cdot \text{vol}(S) \cdot \alpha'_S(\lambda) = \dots = \text{Var}(|X|) > 0. \quad \square$$

Def: expected free volume of the hard sphere model on S :

$$FV_S := \frac{1}{\text{vol}(S)} \int_S \mathbb{P}(d(x_0, X) > 2r_d) dx_0$$

This is the expected fraction of vol. at which a new sphere can be added to X .



Lem S bdd mens. $\subseteq \mathbb{R}^d$ w/ positive vols.

$$\Rightarrow \alpha_S(\lambda) = \lambda \cdot FV_S$$

Pf:
$$\alpha_S(\lambda) = \frac{\mathbb{E}|X|}{\text{vol}(S)} = \frac{1}{\text{vol}(S)} \sum_{k=0}^{\infty} (k+1) \mathbb{P}(|X| \geq k+1)$$

$$= \frac{1}{\text{vol}(S)} \sum_{k=0}^{\infty} (k+1) \frac{\lambda^{k+1} \hat{Z}_S(k+1)}{Z_S(\lambda)}$$

(cf. ① way of counting $\mathbb{E}|I|$ in hard core)

$$\mathbb{E}|I| = \sum_{v \in V(G)} \mathbb{P}_\lambda(v \in I)$$

$$= \frac{1}{\text{vol}(S) \cdot Z_S(\lambda)} \sum_{k=0}^{\infty} \int_{S^{k+1}} \frac{\lambda^{k+1}}{k!} \mathbb{1}_{\mathcal{D}(x_0, x_1, \dots, x_k)} dx_1 \dots dx_k dx_0$$

$$= \frac{\lambda}{\text{vol}(S) Z_S(\lambda)} \int_S \left[\left(1 + \sum_{k=1}^{\infty} \int_{S^k} \frac{\lambda^k}{k!} \mathbb{1}_{\mathcal{D}(x_0, x_1, \dots, x_k)} dx_1 \dots dx_k \right) dx_0 \right]$$

$$= \lambda \cdot FV_S$$



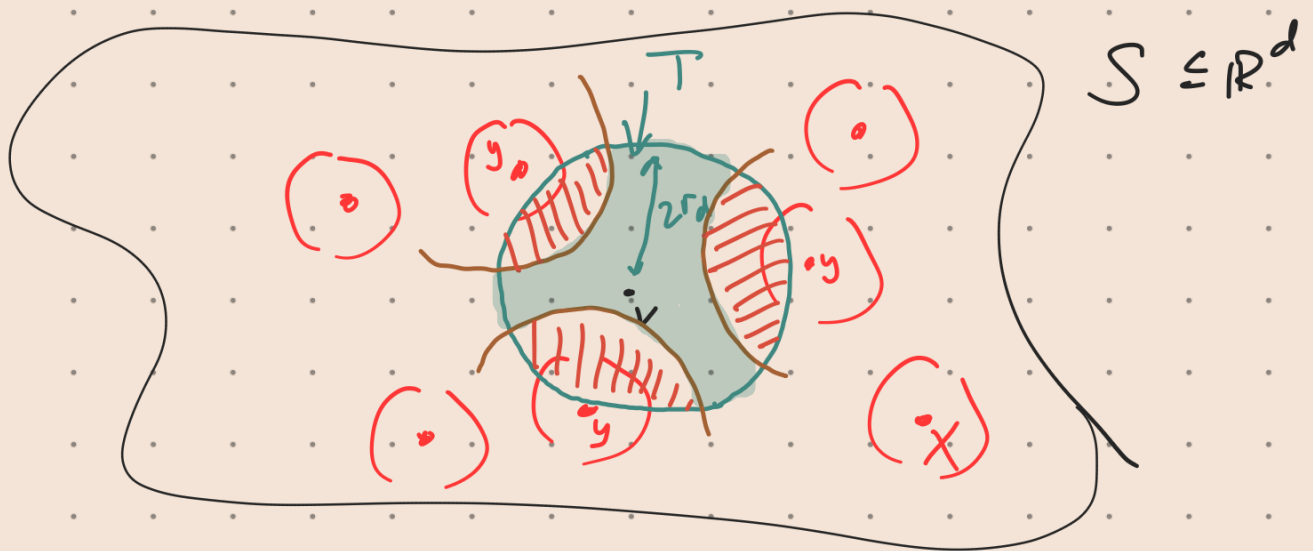
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$$\int_S P(d(x_0, X) > 2r_d) dx_0$$

Experiment :

1. $X \sim$ hard sphere over S w/ fug. λ
2. sample unif pt. v over S

Def : $T = \left\{ x \in B_{2r_d}(v) \cap S : d(x, y) > 2r_d, \forall y \in X \cap \underline{B_{2r_d}^c(v)} \right\}$



• Call T the set of externally uncovered pts.

Idea :

bound $\alpha_S(\lambda)$ in two ways

- 1) use FV_S
- 2) run separate Poisson^{pt} process on T